A Decentralized Monetary Economy with Delayed Production, Employment, and Education

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Abstract

We specify a discrete-time, spatial agent-based monetary economy intended as a reproducible foundation for an economic simulation game. Households value nutritional, residential, material, and social services with nonhomothetic satiating preferences. Firms transform dated inputs and skill-specific labor into outputs through capacity-constrained recipes. Local markets adjust positive price quotes using funded orders and committed supply; agents possess neither the global economic state nor a counterfactual economy simulator. Employment, housing, and education are allocated through local contracts and matching, with bounded preferences for occupational and environmental attributes. Explicit settlement, reservation, and production rules prevent double spending and premature output. We establish accounting feasibility, conservation of money in the closed reference economy, information locality, and nominal scale equivariance. A local linear stability criterion and a separate convex benchmark identify the limited settings in which convergence can be analyzed. No global convergence or empirical calibration of the complete economy is asserted. Appendices provide a 34-good industrial catalog, production coefficients, utility parameters, and building characteristics.

1 Introduction and scope

The desired simulation combines heterogeneous household needs, material production chains, spatial access, employment, and education without assigning intrinsic monetary values to commodities. Prices must be generated by transactions and local adjustment, while preferences and technologies remain physical or behavioral primitives. The objective is a precise disequilibrium mechanism, not an algorithm that presupposes its own equilibrium solution.

Decentralized exchange and the emergence of prices have been studied in agent-based general-equilibrium models [6]. The Eurace@Unibi program connects spatial production, consumption, labor, and human-capital mechanisms [7]. These precedents motivate the architecture, but their numerical findings do not establish properties of the present rules. Classical instability examples also show why an excess-demand price rule cannot be assumed to converge for arbitrary preferences [8]. Recent work studies price adjustment with forward-looking behavior and explicit conditions for stability [9]; those assumptions differ from the adaptive information structure used here.

This article defines a reference algorithm and distinguishes exact invariants from conditional analytical results. The industry appendix is a hypothetical game economy, not an engineering calibration. Public finance is included as a cash-constrained institutional actor. Banking, unsecured credit, endogenous crime, births, deaths, migration, and an unmodeled external buyer are excluded. Initial assets, knowledge, monetary balances, and physical reserves are exogenous endowments. They are never automatically replenished.

1.1 Purpose, design concepts and documentation

This is a model-specification article organized with an ODD crosswalk [1]. Purpose and intended patterns are in this introduction; entities, state and scheduling are in Sec. 2; adaptation, objectives, sensing, interaction and learning are specified in Secs. 3–6. Emergent outcomes are prices, employment, consumption and skill allocation. Stochasticity consists of fixed taste draws, review offsets, tie priorities, illness and injury. Households are collectives with explicit bargaining weights. Initialization, input data and observations are in Sec. 9 and the appendices. No fitted empirical pattern or full dynamic validation is claimed; the static check tests only resource feasibility.

2 State, chronology, and information

2.1 Entities and units

Let $t \in \{0, 1, \ldots\}$ index days. Let $\mathscr{H}$, $\mathscr{I}$, $\mathscr{F}$, $\mathscr{L}$, and $\mathscr{G}$ denote households, persons, establishments, locations, and commodity classes. Each person belongs to one household. Establishments include factories, farms, landlords, schools, retailers, service providers, and transport operators. The government is an additional account with an explicit policy rule. Age in days increases by one each day. Children aged 6–17 may attend basic education; adult remedial students are also eligible. Persons below 18 cannot accept paid employment. At 18 they retain their household, assets and attained skills and become eligible for work and adult courses. Consumption equivalents are 0.5 below age 6, 0.75 from 6 to 17, and 1 thereafter; clothing and occupancy still count individual persons. School occupies five of a child's

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fourteen available daily hours; there is no child productive labor. Births, deaths and migration are disabled, so the population is closed but its age composition changes. No additional age productivity effect or retirement is modeled.

A commodity market $m = (\ell, g, q)$ identifies a location, a good, and a verifiable grade q, including any relevant freshness guarantee. Goods within one market are interchangeable for settlement. Different grades may have different quotes. Let $p_{m,t} > 0$ be the posted quote. Money is the unit of account; no commodity price is fixed as a numeraire. A market is a clearing institution, not a shop building. A shop owns inventory and submits supply into the market, like any other seller; its wholesale purchases and retail sales occur in different dated rounds. The shop provides optional staffed access with a separately posted handling-service fee; direct pickup from a known producer remains possible. The handling fee has its own funded-demand/capacity update and covers shop payroll and paper. Same-grade goods in one market retain one common goods quote; handling and freight charges are separately itemized.

Most material quantities are divisible lots. Jobs, homes, people, school places, and production slots are discrete. The baseline therefore includes indivisibilities, even though a fixed-choice consumption subproblem may be convex. Electricity and dated service capacity are flows, not inventories. Appendix A specifies units and storage rules.

2.2 Complete state

The simulator state X_t contains all variables needed for a Markov transition: cash accounts; cash and inventory reservations; inventories by good, owner, location, and age; durable condition; assets and ownership; jobs and leases; skills, course progress, health, and service stocks; production and construction pipelines; transport queues; natural stocks; price quotes; local beliefs; search sets; and contract histories.

An agent a observes a filtration $\mathscr{J}_{a,t}$ generated by its own state, its own transactions, received messages, visited places, known quotes, and local market announcements. A market sees only the orders, reservations, and settlement records needed for its own clearing. Agents may observe the total orders and offered supply in markets they attend; this is local information, not the state of the economy.

Policies satisfy

$$A_{a,t} = \pi_a(\mathscr{J}_{a,t}, \xi_{a,t}), \tag{1}$$

where $\xi_{a,t}$ is a seeded private or matching random variable. A public research dashboard may inspect X_t, but its aggregates are not automatically passed to π_a. Comparing an agent's own bundles, homes, or job offers is permitted; evaluating alternate economy-wide equilibria is not.

2.3 Reference daily event order

The following order defines the baseline transition, including its otherwise economically consequential timing conventions.

1. Age preexisting lots and remove expired ones to a waste ledger. Then release eligible production from the previous day as new age-zero lots and complete freight. Freight retains the original production age; delivery never refreshes a lot. Apply contractually due transfers that can be funded.

2. On staggered review dates, update local search, job offers, leases, and school applications. Accepted changes take effect on the next day. Every person has one feasible daily activity schedule.

3. For today's existing contracts, reserve payroll, rent, and tuition. Unfunded work or paid service is suspended before it is performed. Reserve the physical inputs and staff needed to offer current utility and service capacity.

4. Households and firms submit jointly affordable orders. Reserve their cash and sellers' physical offers across all markets. Clear each local market once, using the frozen beginning-of-round information.

5. Execute feasible work, transport, service appointments, and production progress. Start new batches only with secured material inputs and current first-stage labor capacity. Release earned payroll and payment for performed services from escrow.

6. Households consume available goods, use durables, and update health and social states. Update skills, condition, forecasts, and financial accounts. Release unused reservations; update tomorrow's quotes from the submitted orders recorded before rationing.

Newly received market revenue cannot finance another order in the same clearing round. End-of-day wages become spendable on the next day. This is a working-capital convention, not a universal fact about money. An implementation with more rounds must retain reservations and use an adjustment rate consistent with its round length.

Daily quantities enforce time and aggregate capacity, not a minute-by-minute transport simulation. Service sessions have explicit time windows; incompatible appointments cannot both be accepted. Employed workers search on a staggered seven-day cycle; unemployed or newly suspended workers search daily. Households review their residence and education on staggered 30-day cycles, with daily housing search when unhoused. Firm offer review occurs with its seven-day production review. The reference production lag is at least one daily boundary for material outputs. Kitchen output becomes available at the next boundary, fresh for that following day. Electricity and other contemporaneous services use reserved delivery capacity instead.

3 Household services and preferences

3.1 Satiation and material attributes

For $z \geq 0$, define

$$H(z) = \frac{3\min\{z, 1/2\} + \min\{(z - 1/2)_+, 1/2\}}{2}. \tag{2}$$

It is increasing, concave, and constant after $z = 1$. The first half of a need has three times the marginal utility of the second half. For a consumption vector x_i, nutritional and sensory services are

$$(N_i, P_i, M_i, Q_i) = \sum_g (n_g, \pi_g, \mu_g, \zeta_{ig} q_g) x_{ig}, \tag{3}$$

with $\zeta_{ig} \in [0.85, 1.15]$ a stable taste multiplier. Nutrition is unaffected by taste. Daily targets are $N_i = 1$, $P_i = 0.5$, and $M_i = 0.5$ per adult; planned consumption obeys $N_i \leq 1.10$. Appendix tables give the attributes and preparation requirements.

The person utility is

$$u_{i,t} = \sum_k w_{ik} H(z_{ik,t}) + 10 \log(1 + L_{i,t}/2) + 30 \log \frac{h_{i,t} + 5}{105} - C^{\text{work}}_{i,t} + P^{\text{home}}_{i,t} + P^{\text{visit}}_{i,t}. \tag{4}$$

Here $h \in [0, 100]$ is health and $L \geq 0$ is discretionary leisure. The household objective uses explicit nonnegative bargaining weights ω_{ih}, with $\sum_{i \in h} \omega_{ih} = 1$. Equal weights are the baseline. Child consumption-equivalence factors alter physical targets, not interpersonal moral worth.

Each person selects at most one clothing set per day. A set with positive condition gives one unit of coverage; condition measures remaining life, not the fraction of the body covered. Fine clothing additionally supplies style proportional to condition. Only the selected set wears; at zero condition it is retired. Furnishing and housing service can remain condition-weighted.

Home hygiene is the minimum of normalized washing water, soap, and facility access. A purchased bath adds delivered hygiene service; its provider has already purchased the water and soap. The customer must not be charged the same physical inputs a second time. Medical care affects expected health transitions rather than adding utility in proportion to money spent.

For k in social connection, recreation, reading, and religious practice,

$$s_{ik,t+1} = \frac{6}{7} s_{ik,t} + \sum_v q_{v,t} a_{kv} y_{iv,t}. \tag{5}$$

Here $y_{iv,t}$ is visits for a paid venue (one hospital unit is a patient-day), and hours for an unpaid activity. Coefficients a_{kv} are service per corresponding unit; visit duration separately enters the time budget. Library sessions are one hour, so their input coefficients per user-hour coincide with per-session coefficients. Stocks above one are retained and decay normally, although their current utility is satiated. A repeated book has declining novelty. Simultaneous borrowing of the same physical copy is prohibited.

The working-age time constraint is

$$L_i + h^{\text{paid}}_i + h^{\text{study}}_i + h^{\text{travel}}_i + h^{\text{domestic}}_i + h^{\text{visits}}_i \leq 14. \tag{6}$$

Eight additional hours are reserved for sleep and two for eating and unavoidable personal care. Queues and searching use actual time. The model cannot satisfy thirty daily food requirements by consuming a monthly quantity on the first day.

3.2 Small, stable attribute preferences

For an option o with d normalized attributes $a_o \in [-1, 1]^d$, set

$$P_i(o) = 2 \tanh\left(\frac{\theta_i^\top a_o}{\sqrt{d}}\right), \qquad \theta_{ik} \sim \text{Uniform}[-1, 1]. \tag{7}$$

Draw separate vectors for occupations, homes, and venues once at initialization. Features include outdoor activity, craftsmanship, helping others, quietness, garden access, density, and familiarity. The total venue-match term is clipped to $[-2, 2]$ per day after weighting each visit by $\min\{1, \text{visit hours}\}$. Reentering a preferred venue therefore cannot generate unlimited utility.

Rent, wages, crowding, pollution, commuting, and physical safety are handled in budgets, services, time, and work costs. They must not be hidden a second time in the small preference bonus. Interest and skill are separate variables. Multiplying a decision maker's entire objective by a common positive number changes no choice. Within a household, rescaling only one member's utility requires an inverse adjustment of that member's bargaining weight to preserve collective preferences. Changing only a taste coefficient does change preferences.

3.3 Local household decision problem

Let $\mathscr{C}_{h,t}$ contain the currently known feasible combinations of home, job, course, and appointment choices. Search is bounded: retain up to five known nearby suppliers per category and discover up to two new local listings weekly, sampled from a public local directory. Never infer an unobserved vacancy or supplier from the global state.

For each $c \in \mathscr{C}_{h,t}$, household h solves its own finite-horizon problem

$$\max_{a_0, \ldots, a_{T-1}} \sum_{\tau=0}^{T-1} \beta^\tau \widehat{\mathscr{U}_{h,t+\tau}}(a_\tau \mid c) + \beta^T \Psi_{h,t}(X_{h,T}), \tag{8}$$

subject to inventory, time, qualification, occupancy, and cash constraints. $\mathscr{U}_h = \sum_{i \in h} \omega_{ih} u_i$ and $\beta = 0.9999$. Use $T = 30$ for routine shopping, $T = 90$ when evaluating a move or job change, and T equal to remaining course duration plus 360 days when evaluating education. These are local planning horizons, not perfect foresight. Execute only the current-day decision and replan tomorrow.

Forecast quotes are the latest observed quotes, held constant. Existing contract incomes are forecast at observed payroll reliability; prospective post-education income uses observed qualified vacancies and the local employment frequency described below. Purchases are forecast at the agent's observed fill fraction. All actual executions use realized fills, not these forecasts. Expected future earnings are never current cash. The baseline evaluates utility along these forecast mean paths: it is a certainty-equivalent planning rule, not an assertion that utility of a mean outcome equals expected utility. Any risk-sensitive alternative must replace this forecast convention explicitly.

For specificity, let the terminal continuation be

$$\Psi_{h,t} = \lambda_{h,t} W^*_{h,t} \log(1 + W_{h,T}/W^*_{h,t}),$$
$$W_{h,T} = m_{h,T} + \sum_g \chi_g \hat{p}_{hg,t} I_{hg,T}. \tag{9}$$

where $W^*_{h,t} = 30 c^{\text{ess}}_{h,t} > 0$ and usable remaining durables enter condition-weighted I, expired goods have value zero, and $\chi_g = 1/2$ is a conservative resale belief. Only goods with known resale outlets enter the sum. This is a private terminal approximation, not a guaranteed redemption facility or an intrinsic price. Let $V_{h,t}(b)$ be the household's one-day attainable utility at current known offers with its discrete arrangements fixed. Define

$$\lambda_{h,t} = \frac{V_{h,t}(B_{h,t} + \delta b) - V_{h,t}(B_{h,t})}{\delta b}, \quad \delta b = 0.01\, c^{\text{ess}}_{h,t}. \tag{10}$$

Here $c^{\text{ess}} > 0$ is the cost of a fixed physical reference basket at known positive quotes, including emergency last-observed quotes when a seller is temporarily unavailable. This basket defines a local monetary scale only; it does not create an order or fix prices. λ may be zero when all attainable one-day services are satiated. Terminal approximations can affect saving and are part of the model's sensitivity analysis.

Today's discretionary budget, after funded fixed obligations, is

$$B_{h,t} = \min\{m^{\text{free}}_{h,t}, [\hat{y}_{h,t} + \kappa_h (m^{\text{free}}_{h,t} - m^*_{h,t})]_+\}, \tag{11}$$

where $\kappa_h = 0.1$, $m^*_{h,t} = 7\hat{e}^{\text{ess}}_{h,t}$, and income and essential expenditures are exponential averages with gain 0.1. The same rule is used in local forecast periods. Initial histories are scenario data. Purchases must fit current free cash even when Eq. (8) forecasts a profitable investment.

Equation (8) defines a preference criterion. Its operational policy uses bounded candidate evaluation, not exhaustive dynamic search. Daily shopping fixes home, jobs, course and durable assignments and solves the one-day continuous problem with the concave terminal-wealth term. For each scheduled discrete review, evaluate the status quo and at most ten alternatives, changing only one arrangement or one member's job. For each candidate, simulate the stated horizon by repeating this one-day policy using frozen quotes and forecast fills, updating stocks, cash, course progress and contract expiry. Choose the best discounted rollout. This is a stationary-policy rollout, not an exact finite-horizon optimum. Use piecewise-linear interpolation of logarithms on a fixed dimensionless grid of 65 equally spaced points over the feasible argument interval; all other satiation pieces are exact. The daily fixed-choice problem is an LP when health effects are held at their current forecast. Long-horizon rollouts occur only on review dates. Ties prefer the current arrangement, then lower spending, then stable identifiers. Report LP count, runtime and objective approximation error in computational experiments.

Proposition 1 (Restricted consumption concavity)**.** *Fix discrete choices, activity schedules, wear assignments, and expected health effects. Suppose remaining service transitions are affine or concave nonnegative minima of affine resources, and use divisible quantities. Then the one-period continuous utility maximization problem with linear resource constraints is a convex optimization problem (maximization of a concave function). With a piecewise-linear leisure approximation it admits a linear-program representation.*

Proof. Equation (2) is nondecreasing and concave; composition with each stated concave service map preserves concavity. Leisure utility is concave in an affine nonnegative residual. Fixed match terms are constants and linear resource constraints define a convex set. For the linear-program form, use two service segments, each bounded by $1/2$, with slopes $3w_k/2$ and $w_k/2$ and total segment use bounded by the available service. Positive decreasing slopes fill the first segment before the second. Standard convex-optimization conventions apply [10]. □

Discrete investments, illness probabilities, and the complete dynamic model are not covered by this proposition.

4 Production, procurement, and investment

4.1 Physical production pipelines

A recipe r has material input vector a_r, output vector b_r, skill-specific effective labor requirements L_{kr}, electricity requirement E_r, and minimum elapsed duration $\tau_r \geq 1$. Starting batch j immediately locks and consumes a_r and occupies one compatible slot. Let $v_{j,t} \in [0,1]$ be accumulated progress. Its daily increment is

$$\Delta v_{j,t} = \min\left\{1 - v_{j,t}, \tau_r^{-1}, \min_{k: L_{kr} > 0} \frac{H_{jk,t}}{L_{kr}}, \frac{e_{j,t}}{E_r}\right\}, \tag{12}$$

omitting the last ratio when $E_r = 0$. $H_{jk,t}$ and $e_{j,t}$ are allocated effective hours and delivered electricity. Only resources corresponding to the chosen increment are used; unused allocations may serve another task. The per-day cap preserves minimum process duration. Output b_r becomes available at the next boundary after $v_j = 1$. Materials are never purchased or consumed a second time after a pause.

Required site resources, installed equipment, and worker qualifications are additional feasibility conditions. Within a firm allocate daily capacity first to contractual service obligations, then to unfinished batches by oldest start date, then to new batches by recipe identifier. Workers' actual hours are disjoint across tasks. General labor contributes effective hours according to skill productivity, but paid hours and time cost remain actual hours. Productivity is defined by Eq. (29).

Services and electricity cannot be sold from future inventory. A provider offers only capacity backed by current inputs, reserved payroll, and feasible schedules. Consumers pay into escrow; providers receive payment only for performed units. The reference power network has no storage and enforces daily delivered energy capacity; finer network dispatch is an extension. A non-electric well and initial water/fuel stocks permit bootstrapping the water–electricity cycle.

4.2 Contemporaneous power capacity

Power is a flow activity, not a delayed recipe. Let $z_{f,t}$ be operating ten-unit blocks, continuously divisible. With installed condition c_f,

$$\begin{aligned} 0 \le z_{f,t} &\le 12c_f, \\ 2z_{f,t} &\le I_f^{\text{charcoal}}, \\ 0.2z_{f,t} &\le I_f^{\text{water}}, \\ (4/3)z_{f,t} &\le H_f^{\text{technical}}, \\ (8/3)z_{f,t} &\le H_f^{\text{general}}. \end{aligned}$$

Delivered generation is $10z_{f,t}$. The six-worker cap and actual-hour schedules apply separately. Reserve inputs and payroll before offering capacity; choose realized z equal to funded delivered demand divided by ten, debit proportional inputs, and release unused reservations. Unserved electricity is never banked. Non-electric wells instead produce one storable water lot per four actual hours, released at the next boundary, with source capacity specified by the scenario.

4.3 Sales beliefs and output release

Each firm updates local forecasts with gain $\alpha = 0.1$:

$$\hat{z}_{f,t+1} = (1-\alpha)\hat{z}_{f,t} + \alpha z_{f,t}^{\text{obs}}. \tag{13}$$

For demand, z^{obs} includes unfilled funded orders, not merely sales. In a common market, assign attempted orders for forecasting by

$$D_{fm,t} = \sigma_{fm,t} D_{m,t}, \qquad \sigma_{fm,t} = \frac{q_{fm,t-1}^{\text{sold}} + \nu C_{fm}}{\sum_{j \in m}(q_{jm,t-1}^{\text{sold}} + \nu C_{jm})}, \tag{14}$$

where C_{fm} is the registered physical daily output capacity and $\nu = 0.05$. This is an observable local demand allocation used for forecasting, not a priority right in actual clearing. If the denominator is zero, no firm is assigned an order. Entry requires an actual advertised technology and financing.

Let I_{fg}^{sell} be saleable output stock after existing commitments. The firm keeps a target buffer $I_{fg}^{*} = b_g \hat{d}_{fg}$, where $b_g = \min\{3, \text{shelf life}/2\}$ days, with $b_g = 3$ for nonperishables. Its offered quantity is

$$S_{fg,t} = \min\{I_{fg,t}^{\text{sell}}, [\hat{d}_{fg,t} + \gamma(I_{fg,t}^{\text{sell}} - I_{fg,t}^{*})]_+\}, \tag{15}$$

with $\gamma = 0.25$. Lots expiring at the next boundary are offered in full, provided they meet the market's stated grade. A plant need not dump its complete buffer each day; otherwise an inventory target can create permanent artificial excess supply.

4.4 A specified local planning policy

On its staggered seven-day review a firm uses

$$\begin{aligned} T_f = \max\big\{31,\ 7 + \max_{r \in f} \tau_r, \\ 1 + \text{longest remaining contract term}\big\}. \end{aligned} \tag{16}$$

An accepted offer includes one paid onboarding day and thirty productive days, hence all thirty-one days enter the hiring comparison. Existing workers beyond expiry are forecast available only under an explicit renewal decision at the current offer, with their observed acceptance frequency; an infeasible renewal generates no labor.

Plans use a time-indexed mixed-integer linear program (MILP): binary batch-start variables for each slot and day; continuous progress, input orders, inventories, output sales, cash and labor allocations; and binary completion indicators. Progress obeys Eq. (12) as linear upper bounds with input debits at start, completion allowed only after unit accumulated progress, and output released one boundary later. Finite slot, storage and cash bounds provide finite indicator bounds. Inputs are separated into age cohorts, expired lots are removed, and sales are bounded by forecast demand and available output. Actual-hour labor is bounded by funded contracts, with fixed known skill multipliers. Future quotes and fill fractions are frozen private beliefs.

The objective for both operation and investment is

$$\begin{aligned} J_f = \sum_{\tau=0}^{T_f-1} \beta^{\tau} \big(&\widehat{\text{receipts}}_{\tau} \\ &- \widehat{\text{payments}}_{\tau}\big) + \beta^{T_f} V_f^{\text{terminal}}, \\ \beta &= 0.9999. \end{aligned} \tag{17}$$

Cash feasibility is enforced separately each day; terminal cash is not added again. Terminal inventories use half the last known resale quote. For unfinished batch j with progress v_j, assign

$$V_j^{\text{WIP}} = \left[\tfrac{1}{2} \sum_g \hat{p}_g b_{rg} - (1 - v_j) \widehat{C}_r^{\text{remaining}} \right]_+ .$$

Remaining cost includes remaining labor and electricity, but excludes already consumed inputs. Assign zero if no known feasible completion route or resale outlet exists. This is a private accounting approximation, not a liquidation guarantee. To avoid speculative terminal accumulation, new starts must have a forecast sales outlet within one recipe duration after the horizon, and total planned sales plus terminal output cannot exceed demand over that extended interval. Existing capital salvage is condition-weighted at known resale quotes; no outlet means zero. Investment uses the identical objective with a 360-day horizon.

Terminal positive-part valuations use bounded indicator variables in the MILP. The MILP search stops after 10,000 branch-and-bound nodes or a relative gap of 1%, whichever occurs first. Execute only a feasible incumbent; the all-idle plan with already committed bills is the fallback if feasible, otherwise suspend unfunded activities. Record the incumbent, bound, relative gap, solver version and deterministic settings. No optimality or game-scale runtime is claimed for this bounded algorithm.

Let R_{fg} be the material requirement of the selected near-term plan, J_{fg} committed inbound deliveries, and I_{fg}^{input} usable input stock. Procurement targets are

$$q_{fg}^{\text{target}} = [R_{fg} - I_{fg}^{\text{input}} - J_{fg}]_+. \tag{18}$$

Define the procurement envelope after today's escrow reservations by

$$B^{\text{proc}}_{f,t} = [m^{\text{free}}_{f,t} - \widehat{K}^{\text{fixed}}_{f,t+1:t+7}]_+,$$

where $\widehat{K}^{\text{fixed}}$ is the undiscounted payroll, rent, tuition, contractual debt-free service bills and essential maintenance over the next seven days, excluding today's already reserved amounts and all discretionary input purchases. Expected receipts do not enlarge this cash envelope. Orders obey $\sum_g \hat{p}^{\text{delivered}}_{fg} q_{fg} \le B^{\text{proc}}_{f,t}$. When the complete target is unaffordable, allocate the envelope by progressive filling of input coverage

$$c_{fg} = \frac{I^{\text{input}}_{fg} + J_{fg} + q_{fg}}{R_{fg}} \quad (R_{fg} > 0). \tag{19}$$

Raise the smallest coverage first, up to one, using known delivered quotes; ties split expenditure to equalize coverage. This is a finite water-filling procedure, not independent spending of the entire budget on every input. It does not guarantee that independent market rationing delivers a complete bundle. Incomplete bundles remain real inventories and affect the next plan. Canceling a project does not refund consumed inputs or undo completed trades.

Each plan implies required worker-hours by skill. Firms announce the unfilled portion as vacancies, subject to their maximum profitable wage and ability to reserve the next day's payroll. For the maximum wage, compare a frozen feasible without-hire incumbent with a with-hire plan over the same horizon, including all thirty-one paid days and onboarding. Search a wage grid with spacing $0.01 c^{\text{ess}}_{\text{adult,t}}/8$ per hour, up to the no-wage gross-receipt plus salvage bound divided by discounted committed hours. Choose the largest grid wage whose feasible with-hire incumbent has nonnegative incremental discounted value and a feasible cash path. This defines a conservative, solver-dependent reservation wage, not an exact global profit threshold. Reserve next-day payroll before acceptance. Subsequent inability to pay invokes the explicit suspension rule; the model does not describe these contracts as fully prepaid guarantees.

Every 30 days an owner may consider one additional building module from its known catalog. For each candidate, evaluate a 360-day cash forecast at observed quotes, include construction time and all bills of materials, and add the same half-quote terminal salvage convention. Start only if incremental discounted value is positive and initial commitments can be funded without consuming reserved payroll. Choose the largest positive value; ties use stable module identifiers. The default discount factor is 0.9999 per day. A household may found a firm by the same rule using its unreserved savings. A 30-day idle firm with insufficient maintenance funds enters liquidation, offering inventories and reusable assets in their actual markets. Sale receipts belong to its owners after funded contractual obligations. No bailout is automatic.

5 Local exchange and price formation

5.1 Reservations and actual settlement

Let $m^{\text{free}}_{a,t}$ be cash after preexisting reservations. A buyer's orders across all attended markets satisfy

$$\sum_m p_{m,t} d_{am,t} + \text{reserved delivery charges} \le m^{\text{free}}_{a,t}. \tag{20}$$

Household discretionary orders also satisfy Eq. (11). Sellers reserve particular inventory lots or dated capacities. The reservation ledger is account-local, allowing atomic rejection of a duplicate commitment without revealing the rest of the economy.

For a divisible material market, define

$$D_m = \sum_a d_{am}, \qquad S_m = \sum_f s_{fm}, \qquad Q_m = \min\{D_m, S_m\}. \tag{21}$$

Executed quantities are

$$x_{am} = d_{am} \min\{1, S_m/D_m\}, \tag{22}$$

$$y_{fm} = s_{fm} \min\{1, D_m/S_m\}, \tag{23}$$

with zero execution if the relevant total is zero. All trades in the round settle at the posted p_m. Transfer $p_m Q_m$ from buyers to sellers, transfer ownership of Q_m goods, and release unused cash and inventory reservations. Select seller lots by earliest expiry, then lot identifier. Goods at another location require separately funded freight; they are unavailable for use until delivered.

Nondivisible homes, jobs, and appointments use the matching procedures below; their allocation must not be implemented as fractions of persons. Services that fail to be delivered release the corresponding escrow to the customer. Failed transport leaves goods with their current owner at the documented location; it does not duplicate them.

Households face the same complementary-input problem as firms. Orders are formed from complete intended cooking bundles net of usable stocks, but proportional fills need not preserve these bundles. After clearing, solve the fixed-choice consumption LP again against actual stocks with no further purchases; consume only recipes whose full food, fuel, water and time inputs are available. Preserve unmatched inputs until used or expired, and record unmet nutrition and stranded inventory. Ready-to-eat alternatives compete in the ex-ante shopping problem. The mechanism guarantees feasible consumption, not satisfaction of planned nutrition.

5.2 Price update

At the end of the round, record funded orders *before rationing* and committed supply *before execution*. Let

$$e_{m,t} = \frac{D_{m,t} - S_{m,t}}{D_{m,t} + S_{m,t} + \varepsilon_m}, \tag{24}$$

$$p_{m,t+1} = p_{m,t} \exp(\eta_m e_{m,t}). \tag{25}$$

Set $\varepsilon_m = 0.01$ of the market's physical trading unit and $\eta_m = 0.02$ per daily round in the reference parameterization. The numerical coefficients bound daily log-price changes, so 0.02 means at most about 2.02% per round, not a fixed 2% change. These are quantity regularization and adjustment-speed parameters, not price anchors. If $D_m = S_m = 0$, the quote is unchanged.

Actual purchases always equal actual sales; substituting either for D_m and S_m would erase the imbalance signal. Previous unfilled orders expire and must be funded again before they count as new demand. This is a posted-price non-tâtonnement process: trades occur away from equilibrium and supply is committed within each round. It belongs to the tradition studied by Hahn and Negishi [2], but their specific stability result is not imported into this monetary rationing model. The rule is a market institution. It is neither a demonstrated profit-maximizing seller policy nor a strategy-proof mechanism. Strategic withholding and order manipulation are outside the price-taking baseline.

Remark 1 (Zero prices and nonexistent supply)**.** *Equation* (25) *preserves strictly positive finite quotes over every finite horizon. A free-disposal good can approach a zero price asymptotically; disposal itself may require a paid service. A positive quote does not establish that a good is produced or available. Entry, financing, and physical feasibility remain necessary.*

6 Employment, residence, and education

6.1 Jobs and wages

For a job j requiring h_{ij} actual hours, use

$$C_{ij}^{\text{work}} = \frac{h_{ij}}{8}(10e_{ij} + 6\rho_i r_j - 2a_j - P_i^{\text{job}}(j)), \qquad (26)$$

with effort e, hazard r, autonomy a, and risk aversion $\rho_i \in [0.7, 1.3]$. Job income affects the feasible household plan, not a separate fixed utility per currency unit. Commute time enters Eq. (6); travel fees enter its budget. Unemployment and feasible self-employment are alternatives.

At each hiring review, process household members in a seeded rotating order. At most one member of any household participates in a matching subround; all other members' jobs are held fixed while computing that member's preference list. Run worker-proposing deferred acceptance on these frozen lists, with each firm ranking eligible workers by effective productivity per paid hour and stable identifier within fixed vacancy classes. Commit that subround before reviewing another member, and process each member at most once per review. Previously committed contracts cannot be displaced. Switching costs 3 utility units. An offer accepted on day t starts paid onboarding on $t+1$, with zero productive output; work starts on $t+2$, and the last paid day is $t+31$. The old job is released at the start of $t+1$.

Deferred acceptance terminates on each finite frozen graph [3]. Its subround stability does not imply stability against joint household deviations: couples create linked preferences [5]. Fixed separate vacancy quotas deliberately avoid asserting substitutability for a firm's complete technology [4]. No global stable or efficient allocation is claimed.

For quote adjustment, A_{fk} counts unique qualified first-choice applicants to class k, not repeat proposals during matching, and V_{fk} is its initially announced vacancy count. When $V_{fk} > 0$,

$$w_{fk,t+1}^{\text{offer}} = \min\left\{w_{fk,t}^{\max}, w_{fk,t}^{\text{offer}} \exp\left(\eta_w \frac{V_{fk} - A_{fk}}{V_{fk} + A_{fk} + 1}\right)\right\}. \qquad (27)$$

Use $\eta_w = 0.02$. If $w^{\max} \le 0$, withdraw the vacancy. With no vacancy, leave the offer quote unchanged. Existing 31-day contracts keep their agreed wage until renewal. A firm unable to reserve a contracted day's payroll suspends that shift before work; the failed promise updates its observed payroll reliability and permits immediate search. There is no unpaid forced labor or hidden negative cash balance.

6.2 Homes and leases

Households rank known affordable vacant dwellings using their complete plan, including household size, condition, service access, travel, and Eq. (7). A move incurs 10 one-time utility units, two adult moving hours when available, and actual transport charges. Housing deposits are zero in the reference model; a later refundable deposit must be treated as an asset and an escrow liability.

Use household-proposing deferred acceptance, with landlords ranking applicants by ability to fund the next payment and then stable seeded priority. Each dwelling is assigned to at most one household and obeys its occupancy limit. The quote is daily rent under a 30-day lease, reserved each morning. If payment cannot be funded, the household must select its available fallback accommodation before using the dwelling; tenancy protection is a policy extension requiring an identified payer.

For each local dwelling class with V available units and A funded first-choice applications, update the new-lease quote by

$$R_{t+1} = R_t \exp\left(\eta_R \frac{A - V}{A + V + 1}\right), \qquad \eta_R = 0.01. \qquad (28)$$

Occupied units keep their contracted rent. Distinct qualities are separate classes; a higher quote does not improve a building's physical attributes.

6.3 Skills and learning

Let $s_{ik} \in [0,1]$ describe general, craft, technical, clinical, and teaching skills. A task with required skill $\bar{s}_k > 0$ receives effective hours

$$H_{ik} = h_{ik}(0.6 + 0.6 s_{ik}), \qquad (29)$$

after enforcing its qualification threshold. For multiple essential skills use the minimum productivity multiplier. General tasks use general skill without a credential barrier. The threshold controls eligibility only: skills 0.5, 0.8, and

1 yield 0.90, 1.08, and 1.20 effective hours per actual hour, respectively.

Course progress is attendance times delivered teacher quality times aptitude. Aptitude satisfies $a_i \in [0.85, 1.15]$. For required effective days T_e, progress P_e changes the targeted skill by

$$s_{ik} = s_{ik}^{\text{start}} + \min\{P_e/T_e, 1\}[s_{ek}^{\text{target}} - s_{ik}^{\text{start}}]_+. \tag{30}$$

Completion and assessment confer the credential. Practice outside courses adds $0.0005(h/8)(1-s)$ per day; an unused occupational skill loses $(\log 2)/(10 \cdot 365)s$ daily (approximately a ten-year half-life). General literacy does not decay in the baseline. Apply learning and decay once per day, then clip to $[0, 1]$.

Schools reserve teachers, inputs, and classrooms before offering places. Applicants rank known programs by Eq. (8); funded first-choice applications and available new places adjust new tuition quotes by the rent-style equation with $\eta_T = 0.01$. Student-proposing matching uses eligibility and stable seeded priority, with continuing students occupying their already contracted places. Publicly financed tuition is paid from the government account, not omitted from accounting.

An agent estimates employment after training from observed eligible vacancies and outcomes of known graduates. Let $\hat{\pi}_e$ be a Beta$(1,1)$ success/failure estimate for employment within 30 days of completion, updated only from observed outcomes. Use the median known eligible wage, or the current fallback wage if no eligible quote is known; expected income is the $\hat{\pi}_e$ mixture of qualified and fallback income. Completion probability is another Beta$(1,1)$ estimate. These private forecasts can be wrong and can produce education cycles. No credential carries an assigned monetary premium.

7 Accounts, resources, and institutional closure

All households, firms, public agencies, and escrow accounts have nonnegative cash. With no borrowing or money issue, a transfer of v from a to b changes their balances by $-v$ and $+v$. Taxes, rents, wages, fees, dividends, and relief are such transfers. Ownership shares $\theta_{hf} \geq 0$ sum to one for each firm. A monthly dividend is θ_{hf} times half of cash above a funded seven-day operating reserve, provided no obligation is overdue. Retained cash stays at the firm. Liquidation transfers assets; it does not erase money.

Government policies are an input schedule or a declared local rule. Taxes are collected only from their specified realized base; subsidies and benefits cannot exceed available treasury cash. If a payment cannot be funded, the policy is rationed by a published priority rule. The reference experiment can set every tax and transfer to zero. A change in public policy must not silently create goods, workers, or currency.

For each storable good let I_g count physical lots, including transit and household stocks but excluding installed assets and consumed batch inputs. Over the half-open interval between successive start-of-day states,

$$\begin{aligned} I_{g,t+1} = I_{g,t} &+ B_{g,t}^{\text{complete}} + B_{g,t}^{\text{well}} \\ &- A_{g,t}^{\text{batch}} - A_{g,t}^{\text{venue}} - A_{g,t}^{\text{power}} \\ &- C_{g,t}^{\text{food/home}} - A_{g,t}^{\text{build}} - A_{g,t}^{\text{repair}} \\ &- A_{g,t}^{\text{install}} - W_{g,t}^{\text{retire}} - W_{g,t}^{\text{expiry}}. \end{aligned} \tag{31}$$

Venue inputs include bathhouse water/soap, clinic medicines, school inputs and all delivered-service coefficients. Home use includes food, drinking and washing water, soap and cooking fuel/water. Construction, repair and installation have separate event codes, each debited once. Well output is explicit. Power fuel is debited as delivered flow is produced. Transfers between owners cancel. Durable use changes a separate condition ledger; only retirement removes the physical item from I_g. Installed asset counts, construction progress, condition and retired waste have separate ledgers; condition-weighted value never replaces a lot count. Recipe tool wear is classified as batch input, not retirement or repair. Electricity satisfies delivered generation equals delivered use (zero storage and zero network loss in this baseline) and is excluded from this inventory equation.

Natural-resource regeneration is a scenario parameter. A renewable fish stock may use $Z_{t+1} = Z_t + rZ_t(1 - Z_t/K) - q_t$, with harvest restricted to available stock. Forest cohorts mature after the specified delay. These physical constraints do not assign a price to nature. Land can be privately leased or publicly allocated, but ownership and use must be specified. The closed reference economy has no unlimited export buyer.

8 Analytical properties and their limits

Theorem 1 (Accounting and physical feasibility)**.** *Suppose initial cash and inventories are nonnegative, reservations are atomic, allocations satisfy the stated capacity and time constraints, and every execution is bounded by its reservation. Then the reference algorithm preserves nonnegative cash and inventories, prevents duplicate use of committed goods and time, and releases material output only after its production requirements have been completed. In the closed no-credit economy, total money including escrow is conserved.*

Proof. Proceed by induction through the event order. A reservation reduces spendable resources without changing their total ownership. Since all orders of an account share the same ledger, their costs cannot exceed its free balance. Equations (22)–(23) execute no more than each reserved order, and aggregate purchases and sales both equal Q_m. Each payment is an equal debit and credit, including transfers into and out of escrow. Consumption, expiry, maintenance, and input commitment are bounded by owned available stock. Slots and staff schedules are reserved before execution. Equation (12) cannot exceed remaining progress,

labor, electricity, or the process rate, and completion is required before output release. Refunding unused reservations reverses no completed transaction. Thus each event preserves the induction hypotheses, and no event changes the sum of cash balances. Monetary conservation does not imply conservation of utility, employment, or the quantity of each good. □

Lemma 1 (Finite-horizon price bounds)**.** *For a market with constant $\eta > 0$ and initial $p_0 > 0$,*

$$p_0 e^{-\eta t} \le p_t \le p_0 e^{\eta t}. \tag{32}$$

Proof. The normalized imbalance has absolute value at most one. Sum the logarithmic increments in Eq. (25). □

These bounds prevent a finite-step zero or infinity in exact arithmetic. They give no uniform infinite-horizon bound. Numerical implementations should store log quotes; a numerical underflow guard is not an economic price floor.

Lemma 2 (Information locality)**.** *If initial beliefs are local and independent matching randomness is recorded, every action defined above is measurable with respect to the agent's information and its received local messages. The algorithm requires no counterfactual global production allocation.*

Proof. Household plans use their own states, known candidate sets, and local beliefs. Firms use their own recipes, finance, inventories, workers, and market announcements. Clearing uses submitted orders only. All subsequent messages are functions of these quantities and recorded randomness. Induction on event time establishes the claim. □

Proposition 2 (Nominal scale equivariance)**.** *Multiply initial cash, all quotes and contracts, monetary histories, and monetary policy thresholds by the same $c > 0$. Preserve physical states, dimensionless parameters, and random draws. In exact arithmetic, the algorithm produces the same real allocations, with every monetary quantity multiplied by c.*

Proof. Budget and escrow inequalities scale by c, so their feasible real bundles are unchanged. The essential-basket cost and the finite-difference step in Eq. (10) scale by c, while λ scales by $1/c$; the terminal utility is therefore unchanged. Firm objectives and maximum profitable wages scale by c without changing their ranking. Tie rules use scale-invariant comparisons. Funded quantities, matchings, and physical updates are unchanged. Normalized imbalances are unchanged, so multiplicative price, wage, rent, and tuition updates preserve the scaling. Induct over days. □

Changing the distribution of wealth is not this transformation. Neither path independence nor a unique relative-price vector follows.

8.1 Local price stability

Freeze other states and suppose a differentiable price-only response $e(\ell)$ exists for log quotes $\ell = \log p$. This assumption fails at many discrete switches and must be checked in the particular benchmark. At $e(\ell^*) = 0$, put $J = De(\ell^*)$. For heterogeneous gains the linearization is $I + \mathrm{diag}(\eta_m)J$, and its spectral radius must be less than one. With a common step η,

$$\delta\ell_{t+1} = (I + \eta J)\delta\ell_t + o(\|\delta\ell_t\|). \tag{33}$$

Proposition 3 (Conditional local stability)**.** *If $\rho(I + \eta J) < 1$, the fixed point is locally asymptotically stable. If an eigenvalue has modulus greater than one, it is unstable. Eigenvalues on the unit circle leave the linearization test inconclusive. For $\lambda \in \mathrm{spec}(J)$ with $\mathrm{Re}\,\lambda < 0$, its linear mode is stable precisely when*

$$0 < \eta < \frac{-2\,\mathrm{Re}\,\lambda}{|\lambda|^2}. \tag{34}$$

Proof. The first assertions are the discrete-time linearization criterion. The last follows by expanding $|1 + \eta\lambda|^2 < 1$. □

A positive-real-part eigenvalue cannot be stabilized merely by reducing a positive step. For the actual economy one must augment the Jacobian with inventories, finance, contracts, expectations, and pipelines. The price-only proposition is not a proof for that augmented system.

8.2 A separate convex coordination benchmark

To isolate a provable price mechanism, consider compact convex sets Y_a, concave continuous payoffs f_a, and affine resource use r_a. Let

$$\max_{y_a \in Y_a} \sum_a f_a(y_a) \quad \text{subject to} \quad \sum_a r_a(y_a) \le b. \tag{35}$$

Assume Slater feasibility; compactness makes the primal optimum finite, and Slater's condition gives dual attainment. Interpret f_a as quasilinear net benefit in a common numeraire with nonbinding finance; it is not an arbitrary sum of the original households' cardinal utilities. The dual is

$$\Phi(p) = p^\top b + \sum_a \max_{y_a \in Y_a} \{f_a(y_a) - p^\top r_a(y_a)\}, \quad p \ge 0. \tag{36}$$

Each maximization is private. Resource markets need only summed resource use, consistent with dual decomposition [10, 11].

Theorem 2 (Restricted dual-price convergence)**.** *Under the preceding assumptions and positive steps with $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$, the update*

$$p_{t+1} = \left[p_t + \alpha_t \left(\sum_a r_a(y_{a,t}) - b \right) \right]_+ \tag{37}$$

with exact private maximizers converges to a dual minimizer in finite dimension.

Proof. A subgradient of Φ is $g_t = b - \sum_a r_a(y_{a,t})$. Compactness of the finite product of Y_a and affine resource maps gives $\|g_t\| \le G < \infty$. Nonexpansiveness of projection and the subgradient inequality give, for any dual optimum p^*,

$$\|p_{t+1} - p^*\|^2 \le \|p_t - p^*\|^2 - 2\alpha_t[\Phi(p_t) - \Phi(p^*)] + \alpha_t^2 G^2. \tag{38}$$

Summability of the last term makes distances to the optimum set convergent and the sequence bounded. Since the steps have divergent sum, the nonnegative objective gap has zero liminf. A bounded subsequence attaining that liminf has an optimal cluster point by continuity. Distance convergence to this cluster point then implies convergence of the whole sequence. □

This is an additive projected dual update for a different, restricted problem. It does not prove convergence of the normalized multiplicative cash-constrained algorithm, convergence of every primal allocation, or efficiency of the game's distribution of income.

9 Implementation defaults and validation protocol

Physical catalogs and preferences are scenario inputs, fixed before an experiment. Use at least 35 days of initial basic food, working assets, essential skilled workers, and viable water and fuel access. Test whether crop plots and initial production capacity can physically support the population before attributing starvation to a market rule. Fixed initial assets are necessary to bootstrap circular tool–steel–machinery chains.

Record the simulator version, scenario checksum, random seed, price initialization, wealth distribution, and every parameter. Stagger review dates by a seeded uniform offset, rather than letting every household or firm adjust simultaneously. Default gains and review periods are listed in Appendix E; they are experimental values, not established stabilizing constants.

The required diagnostic suite is: (i) exact cash and reservation checks; (ii) physical flow and age bookkeeping; (iii) nominal rescaling; (iv) consumer utility and substitution checks at fixed quotes; (v) local stability analysis where differentiability applies; and (vi) repeated-seed shock experiments with harvest, fuel, workforce, transport, and schooling disruptions. Report unmet needs, unemployment, vacancies, service queues, stranded complementary inputs, bankruptcies, price dispersion, and recovery times. Failed economies are outcomes to report, not runs to discard.

Compare alternative market rules while holding physical endowments, information, and institutions fixed. A centralized allocation solver may serve as an external benchmark, but its solution must not be passed to agents. Distinguish financial insolvency, physical impossibility, search failure, and coordination failure. A high price alone identifies none of these uniquely.

The reported checks comprise food-processing nutrient inequalities and the static feasibility calculation in Appendix G. No full-economy simulation results, estimated parameters, convergence rates, or empirical fit are reported here.

The reference daily discount gives $\beta^{365} \simeq 0.9642$ and $\beta^{720} \simeq 0.9305$, an annual effective rate of about 3.72%. Goods can adjust every day, new wages every seven days, and new rents and tuition every thirty days; maximal average log-adjustment speeds are therefore 0.02, 0.02/7 and 0.01/30 per day. This is an explicit contractual rigidity hypothesis. It can generate real-wage swings. Sensitivity experiments must also use common per-day speeds, e.g. review gains 7η and 30η, and vary the common η; the reference asymmetry is not a stability calibration.

10 Conclusion

The model specifies a local monetary mechanism through which household needs generate retail orders, firms transmit derived orders upstream, and wages, rents, and education respond to locally observed opportunities. Physical production and service capacity prevent prices from substituting for missing resources. Explicit reservation and settlement establish feasibility independently of whether the economy is prosperous or stable. The main unresolved scientific question is the stability and efficiency of the coupled adaptive system under complementary technologies, financing constraints, and delayed skill supply. The proposed catalog and algorithm make that question reproducible without presuming its answer.

A Goods, units, and storage

The catalog comprises exactly 34 commodity classes. One water unit is 100 liters and one electricity unit is 10 kWh. Other quantities are commodity-specific lots. In particular, a grain lot supplies one adult-day of the energy index before losses; it is not interchangeable with a kilogram of every other good. Equipment and garments are durable lots; fractional orders can represent pooled procurement, while household allocation and physical objects can be made indivisible in a declared extension. No row has a base price.

Goods are tracked by lot age and condition. Shelf-life expiry is an explicit removal to waste, not an unexplained second depreciation charge. Waste remains a recorded physical residual. The reference scenario provides a finite permitted landfill volume as an endowment; once full, waste-producing activity requires funded disposal capacity or stops. No disposal industry or pollution-removal technology is assumed to exist without an explicit recipe. Electricity cannot be stockpiled.

Table 1: Launch commodity catalog and storage conventions.

ID	Commodity	Uses	Storage / durability
G01	Grain	Milling; household staple preparation	180-day shelf life
G02	Legumes	Household food; preserved food; prepared meals	180 days
G03	Vegetables and fruit	Household food; diet quality; food processing	7 days
G04	Fish	Household protein; food processing	2 days without refrigeration
G05	Timber	Boards; charcoal; paper	Dry storage; no spoilage in the reference model
G06	Textile fibre	Fabric	Dry storage
G07	Oilseed	Edible oil	180 days
G08	Iron ore	Steel	No spoilage
G09	Limestone	Cement; steel flux	No spoilage
G10	Clay	Bricks; cement	No spoilage
G11	Salt	Preserving food; soap	Dry storage
G12	Medicinal plants	Medicines	30 days
G13	Flour	Bread; household staple preparation	90 days
G14	Boards	Buildings; furniture; tools; machinery	Dry storage
G15	Fabric	Garments; furniture; medicines	Dry storage
G16	Charcoal	Process heat; cooking; power	Dry storage
G17	Steel	Machinery; tools; construction; food containers	No spoilage
G18	Cement	Construction and maintenance	90 days in dry storage
G19	Bricks	Construction and maintenance	No spoilage
G20	Edible oil	Cooking; meals; soap	90 days
G21	Paper	Books; schools; administration; packaging	Dry storage
G22	Tool kits	Equipment and maintenance inputs	Durability stock, not an edible flow
G23	Machinery modules	Plant capacity and repair	Durability stock
G24	Soap	Household hygiene; clinics; services	365 days
G25	Bread	Convenient staple food	3 days
G26	Preserved food	Durable mixed food	365 days
G27	Prepared meals	Convenient, pleasant mixed food	1 day
G28	Basic clothing	Coverage and basic comfort	180 wear-days at full condition
G29	Fine clothing	Coverage, comfort, and style	240 wear-days
G30	Furniture kits	Household comfort; venue equipment	1,800 occupied days
G31	Books	Reading; libraries; education	180 reading sessions before replacement
G32	Medicines	Treatment input; no direct consumption pleasure	180 days
G33	Clean water	Drinking; hygiene; processing	Stored water lasts 7 days unless treated
G34	Electricity	Machines; lighting; ventilation; services	Instantaneous network service; no storage in the reference model

B Production coefficients and construction

The following tables provide all commodity-production routes in the launch catalog. The hours column is total effective labor per batch; the duration is a minimum process time. Worker and slot limits are separate constraints. Input quantities are game parameters, not measured industrial coefficients. An asterisk identifies replacement/tool wear consumed from inventory; it does not replace the need for installed equipment. Material output is released at the next boundary after completion.

Any required named supervisor must be a qualified member of the listed workforce. For a recipe requiring one or two technicians, allocate one or two equal worker-shares of its labor requirement to technical labor: one share is total recipe hours divided by the declared staffing reference (the worker cap). This fixed labor mix is independent of realized employment. The rest is general labor. A craft-plus-technician machine shop allocates one such share to technical labor and the rest to craft

labor. A medicinal-garden supervisor requires craft skill at least 0.5. This convention makes the labor vectors in the main text numerical.

Table 2: Extraction and agriculture.

Facility	Inputs	Output	h	Lag	Workers / slots	Skills
Grain farm	2 water + 0.01 tools*	20 grain	40	30 days	2 / 12 plots	General
Legume farm	2 water + 0.01 tools*	10 legumes	32	30 days	2 / 12 plots	General
Vegetable farm	4 water + 0.01 tools*	20 vegetables	32	10 days	2 / 4 plots	General
Fibre farm	3 water + 0.01 tools*	10 fibre	32	30 days	2 / 12 plots	General
Oilseed farm	2 water + 0.01 tools*	10 oilseed	32	30 days	2 / 12 plots	General
Medicinal garden	2 water + 0.01 tools*	5 medicinal plants	24	20 days	4 / 6 plots	General; one trained supervisor
Fishing dock	0.02 tools*	10 fish	32	1 day	8 / 2 boats	General
Forestry camp	0.03 tools*	10 timber	24	1 day	8 / 3 cutting slots	General
Iron mine	0.05 tools* + 0.1 electricity	10 ore	40	1 day	12 / 3 slots	General; one technician
Limestone quarry	0.03 tools*	10 limestone	24	1 day	8 / 3 slots	General
Clay pit	0.02 tools*	10 clay	16	1 day	6 / 3 slots	General
Saltworks	Seawater access + 0.01 tools*	5 salt	16	3 days	6 / 9 evaporation beds	General

Table 3: Intermediate industry and utilities.

Facility	Inputs	Output	h	Lag	Workers / slots	Skills
Mill	10 grain + 0.1 electricity	9 flour	8	1 day	4 / 4	General
Sawmill	10 timber + 0.1 electricity + 0.02 tools*	8 boards	12	1 day	6 / 4	General
Textile mill	10 fibre + 1 water + 0.2 electricity	8 fabric	24	1 day	8 / 3	General; one technician
Charcoal kiln	10 timber	4 charcoal	8	2 days	4 / 8	General
Steelworks	10 ore + 5 charcoal + 1 limestone + 1 water + 0.2 electricity	5 steel	40	2 days	12 / 5	General; two technicians
Cement works	8 limestone + 2 clay + 2 charcoal + 0.2 electricity	8 cement	24	2 days	8 / 5	General; one technician
Brickworks	10 clay + 1 water + 2 charcoal	8 bricks	16	2 days	8 / 6	General
Oil press	10 oilseed + 0.1 electricity	2.5 oil	12	1 day	4 / 2	General
Paper mill	4 timber + 4 water + 1 charcoal + 0.2 electricity	8 paper	24	2 days	8 / 5	General; one technician
Tool workshop	2 steel + 1 boards + 0.1 electricity	1 tools	16	1 day	6 / 3	Craft
Machine shop	5 steel + 1 boards + 0.1 tools* + 0.5 electricity	1 machinery	40	3 days	8 / 4	Craft plus technician
Soap works	1 oil + 0.2 salt + 0.5 water + 0.2 charcoal	3 soap	8	1 day	4 / 3	Craft
Water plant	Freshwater access + 0.1 electricity	10 water	2	1 day	4 / 12	One technician
Power station	2 charcoal + 0.2 water	10 electricity	4	Flow	6 / 12 flow units	Two technicians

Table 4: Food processing and household goods.

Facility	Inputs	Output	h	Lag	Workers / slots	Skills
Bakery	10 flour + 3 water + 1 charcoal	14 bread	12	1 day	6 / 4	Craft
Food preservatory	6 fish + 4 legumes + 3 vegetables + 0.5 oil + 0.3 salt + 0.2 steel + 0.2 paper + 1 charcoal + 1 water	10 preserved food	24	1 day	8 / 3	Craft
Kitchen / restaurant	4 bread + 3 legumes + 4 vegetables + 2 fish + 0.5 oil + 1 water + 0.5 charcoal	10 prepared meals	16	1 day	8 / 4 daily batches	Craft
Clothing workshop, basic	2 fabric	1 basic clothing	8	1 day	8 / 8	Craft

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Table 4 (continued)

Facility	Inputs	Output	h	Lag	Workers / slots	Skills
Clothing workshop, fine	3 fabric + 0.02 tools*	1 fine clothing	16	2 days	8 / 8	Advanced craft
Furniture workshop	3 boards + 0.5 fabric + 0.1 steel + 0.02 tools*	1 furniture	16	2 days	8 / 8	Craft
Printer	1 paper + 0.05 oil + 0.05 electricity	1 books	2	1 day	4 / 12	Literate craft
Pharmacy works	1 medicinal plants + 0.2 fabric + 0.1 paper + 0.2 water + 0.05 electricity	5 medicines	8	2 days	4 / 6	Clinical technician

Freshwater, seawater, agricultural plots, fish biomass, forest cohorts, and mineral deposits are physical site resources. Crop yields multiply by fertility in $[0,1]$. Water is an upfront input; a batch lacking its full water requirement cannot start. There is no separate water-completion factor. Crop-farm worker caps are two: at unit productivity a grain farm's 12 plots require $12(40)/30 = 16$ hours daily, exactly two eight-hour workers. Lower skill or absence makes labor binding; faster workers remain limited by plots. Farm labor is entirely general, so no supervisor-share rule is applied to these farms. Farm plots are occupied for the entire growth cycle. Forestry consumes standing timber and regrows by cohorts after 1,800 days. Fish use logistic regeneration with scenario parameters r and K; mineral extraction reduces finite deposits.

A non-electric emergency well yields one clean-water unit per four actual labor-hours at a suitable source. The soap recipe aggregates minor mineral reagents into the salt input; it is not a chemical manufacturing instruction. All unlisted residues are waste, not free additional output.

For construction, letters A–E below denote bills of materials, not tradable commodities or currency-valued building costs. Land and access are additional constraints. A construction worker uses an installed tool allocation of one kit per four workers; wear consumes 0.01 tool kits per 80 actual construction hours.

Table 5: Structural construction packages.

Package	Boards	Bricks	Cement	Steel	h	Days
A: light structure	12	4	0	0	80	4
B: small masonry structure	12	20	8	2	160	8
C: workshop / service building	20	40	16	5	320	12
D: industrial hall	30	80	40	20	640	20
E: apartment block / large institution	40	160	80	40	1,280	30

Facility installation packages are: farms A plus 0.2 machinery modules; medicinal garden A plus 0.1; fishing dock B plus 1; forestry, quarries, clay pits, and saltworks A plus 0.5; mine D plus 3; mill, sawmill, and oil press C plus 2; textile and paper mills D plus 4; kiln and brickworks C plus 1; steel and cement works D plus 6; tools, garments, furniture, soap, bakery, and printer C plus 1; machine shop D plus 4; preservatory D plus 3; pharmacy C plus 2; water and power plants D plus 5. Each industrial establishment also installs at least one tool kit.

Installed machinery requires replacements equal to 0.0005 of installed modules per operating day. Each building, occupied or vacant, requires materials and repair labor equal to 0.0001 of its original construction bill daily. Accumulate fractional repair credits rather than transporting infinitesimal pieces. Let r_t be delivered repair as a fraction of the full original bill. Set daily building depreciation $d = 10^{-4}$, including vacancy; machinery depreciation is $5 \cdot 10^{-4}$ on operating days and zero when mothballed. Update condition as $c_{t+1} = \min\{1, \max\{0, c_t - d\} + r_t\}$; all required repair components must be supplied proportionally. Machinery uses the analogous rule with its own replacement rate. Usable capacity and shelter multiply by condition. Replacements restore condition, not additional installed capacity. Tools explicitly consumed by a recipe are not charged a second maintenance debit for that use.

C Consumption, utility, and health parameters

Food attributes are defined per consumed lot. The adult targets are energy $N = 1$, protein $P = 0.5$, and micronutrients $M = 0.5$; Q is sensory quality. These are simulation indexes, not dietary recommendations.

Table 6: Food-service coefficients.

Food	Energy N	Protein P	Micro. M	Enjoyment Q	Family
Grain	1.00	0.15	0.05	0.05	Staple
Legumes	0.80	0.70	0.20	0.15	Legume
Vegetables / fruit	0.25	0.10	1.00	0.25	Produce

Continued on next page

Table 6 (continued)

Food	Energy N	Protein P	Micro. M	Enjoyment Q	Family
Fish	1.00	1.00	0.30	0.30	Fish
Flour	1.10	0.15	0.02	0.05	Staple
Edible oil	3.00	0	0	0.10	Oil
Bread	0.75	0.10	0.01	0.35	Staple
Preserved food	1.00	0.80	0.40	0.35	Mixed
Prepared meals	0.95	0.45	0.45	1.00	Mixed

Grain, flour, legumes, and raw fish require preparation. Preparing one energy unit uses 0.4 adult-hours, 0.03 charcoal, and 0.01 water, in addition to the food itself. Serving bread or preserved food uses 0.05 hours per energy unit; purchased prepared meals require no further preparation. Eating time is already in personal care. Oil is an ingredient limited to 0.10 lots per adult-day and must accompany other food.

The four food-processing recipes satisfy the componentwise inequalities

$$A^{\text{nutrient}} b_r \le A^{\text{nutrient}} a_r.$$

The output/input energy ratios for milling, baking, preserving, and meal preparation are respectively 0.990, 0.955, 0.873, and 0.960. Enjoyment is not a conserved nutrient and may increase through cooking.

One wearable clothing set covers one adult irrespective of remaining condition. One wear-day removes $1/180$ condition from a basic set or $1/240$ from a fine set. Only one set may be assigned; fine style equals its remaining condition. Furniture loses $1/1800$ condition per occupied day. A book copy lasts 180 reading sessions. Title-specific novelty multiplies reading service by $1, 1/2, 1/4$, then $1/8$ on later readings. Library lending reserves the physical copy.

Table 7: Baseline weights in Eq. (4).

Service	Normalization / interpretation	Weight
Safe drinking water	$d_i/0.03$	120
Energy from food	N_i	100
Weatherproof shelter	occupied home's shelter index	50
Clothing coverage	one assigned wearable set	28
Protein	$P_i/0.5$	18
Micronutrients	$M_i/0.5$	18
Hygiene	home hygiene + purchased bathhouse hygiene	14
Thermal comfort	climate-adjusted comfort index	14
Space	$A_h/(20 n_h^{\text{eq}}\,\text{m}^2)$	8
Privacy	household privacy index	6
Furnishing	effective furniture kits / adult equivalents	5
Clean environment	1 - residential pollution exposure	12
Personal security	1 - local crime exposure	10
Social connection	social stock	10
Recreation	recreation stock	8
Reading / cultural engagement	reading stock	5
Religious practice	religious stock	0 to 8
Food enjoyment	$\sum_g \zeta_{ig} q_g x_{ig}$	6
Clothing style	condition of the assigned fine set	4

Multiply non-survival weights independently by a seeded draw from $[0.85, 1.15]$; water and food-energy weights instead use $[0.95, 1.05]$. Half the initial population has religious weight zero; the remainder draw uniformly from $[2, 8]$ without a further multiplier. These are synthetic heterogeneity assumptions. No food-variety entropy bonus is included.

For a minimal fictional health module, with d_t drinking water in water units, use

$$h_{t+1} = \text{clip}_{[0,100]}\left[h_t + 0.2 - 4(1-N_t)_+ - 8(1-d_t/0.03)_+ - I_t^{\text{loss}} - J_t^{\text{loss}} + T_t^{\text{gain}}\right]. \tag{39}$$

A minor illness lasts seven symptomatic days and causes one health-point loss daily. Only a currently uninfected person draws a new illness, with probability $0.002 + 0.01(1 - \min\{z_{\text{hygiene}}, 1\}) + 0.005$ pollution exposure. One successful clinic visit reduces remaining symptomatic time by $3q$ days with probability 0.8; fractional days reduce the last day's loss proportionally. Same-day repeat treatment has no further effect. The clinic cannot improve health above 100 by selling visits to a healthy person.

Per work-hour, a lost-time injury occurs with probability $2 \times 10^{-6} + 2 \times 10^{-5} r$. Draw one uniform integer absence duration from 3 to 10 days and an immediate 10-point health loss; at most one new injury is drawn per person-day. Injured

persons do not work during their absence. This hazard module is illustrative, not occupational epidemiology. Severe illness and hospital treatment are disabled in the reference stochastic experiment until a treatment transition is supplied; the hospital remains a catalogued service technology for that extension.

Residential pollution is produced by operating batches: 0.05 load for farms and fishing, 0.2 for extraction and light industry, and 1.0 for charcoal, steel, cement, and power. Allocate each load among cells using normalized weights $\exp(-d/1\,\mathrm{km})$, with distance measured from the emitting establishment. Local stock retains 90% daily and exposure is $1-\exp(-\mathrm{stock}/10)$. Services contribute 0.02 per operating establishment-day. These are synthetic environmental units. Crime exposure is a fixed scenario field; endogenous policing and crime are outside the reference model.

D Workplaces, homes, and service venues

An actual vacancy additionally has a location, hourly wage quote, payroll history, required skill vector, schedule, contract, and training access. Effort, hazard, and autonomy in Table 8 are adjustable physical and organizational baselines. Outdoor share and noise are features for tastes or an explicit health extension, not duplicate unexplained effort penalties.

Table 8: Workplace baseline attributes.

Workplace	Effort	Hazard	Autonomy	Outdoor share	h
Crop farm / garden	0.55	0.15	0.45	0.90	8
Fishing / forestry	0.65	0.55	0.50	0.95	8
Mine / quarry	0.80	0.75	0.15	0.40	8
Kiln / steel / cement	0.75	0.60	0.15	0.20	8
Mill / textile / paper	0.50	0.30	0.20	0.10	8
Food processing / bakery	0.45	0.20	0.35	0.10	8
Clothing / furniture / tools	0.40	0.25	0.60	0.10	8
Machine shop / utilities	0.45	0.35	0.50	0.20	8
Shop / hospitality	0.45	0.10	0.35	0.10	8
Freight / construction	0.65	0.45	0.40	0.80	8
Teacher / librarian	0.35	0.05	0.65	0.05	6
Clinical worker / pharmacist	0.50	0.15	0.55	0.05	8
Administration	0.25	0.03	0.40	0.05	8

A dwelling's rent is for the unit, not automatically per resident. Privacy and thermal comfort are indexes in $[0,1]$. Area utility uses adult-equivalents, whereas the hard occupancy limit counts persons. Condition, weather, actual utility delivery, neighborhood exposure, and travel time modify the realized services. A block uses either the eight basic-unit layout or the five family-unit layout, not both.

Table 9: Residential types and construction requirements.

Dwelling	Area	Max. people	Shelter	Temp.	Priv.	Facilities	Package
Makeshift shelter	12 m^2	2	0.45	0.30	0.45	No private water, sanitation, or kitchen	0.25 A
Worker dormitory	80 m^2	8	0.85	0.50	0.20	Shared water, washroom, kitchen	B
Basic apartment	36 m^2	3	0.95	0.65	0.80	Private kitchen/washroom; utility connections	One of 8 units in E
Family apartment	60 m^2	4	1.00	0.80	0.90	Private facilities; optional powered ventilation	One of 5 units in E
Cottage	55 m^2	4	0.90	0.70	0.95	Kitchen/washroom; garden; lower density	B
Comfortable house	90 m^2	5	1.00	0.90	1.00	Private facilities; garden; powered comfort option	C

Connected lighting uses 0.02 electricity units per occupied dwelling-day. Optional powered ventilation uses 0.03 per adult-equivalent/day and can add up to 0.10 thermal comfort when climatically useful, capped at one. Without electricity the powered increment disappears, but structural shelter does not. Garden, density, architectural style, quietness, and known social ties are the small home-affinity features. Unhoused persons have shelter and privacy zero and use only actually available public facilities.

Venue quality is $q = c(0.6+0.4\bar{s})u$, with condition c, relevant mean skill $\bar{s}$, and supplied fraction u, all in $[0,1]$. For complementary quality inputs, u is their minimum supplied/required ratio. Essential safety inputs are hard prerequisites. Capacity limits produce queues or rejected bookings rather than fully served fictitious visitors. Inputs listed below are per visit unless otherwise stated; ordinary building maintenance and installed equipment are additional.

Table 10: Dated service technologies and household time.

Place	Building, staff, capacity	Variable inputs	Time	Delivered service
Market shop	C; 4 general workers; 80 transactions	0.005 paper; inventory acquired separately	0.10 h + queue	Goods and lower shopping/search time; no extra consumption utility
Communal kitchen	C + 1 machinery; 4 craft workers; 20 meal-equivalents	Food and cooking recipe inputs	0.10 h + queue	Access to preparation for homes without kitchens
Restaurant	C + 1 machinery; 8 staff; 30 diners	1 prepared meal + 0.02 water + 0.002 soap	0.75 h + queue	Food nutrients/Q once; social +0.15q; recreation +0.10q
Bathhouse	C + 1 machinery; 4 staff; 40 visits	0.5 water + 0.01 soap + 0.02 electricity	0.40 h + queue	Hygiene for one day at access quality q
Park	Landscaped land + 0.25 A; 2 staff; 100 visitors	Maintenance labor; 0.01 water per visit-equivalent	1 h	Recreation +0.5q; social +0.2q only when meeting others
Social club / tavern	C; 6 staff; 40 visits	0.01 water + 0.002 soap; optional food bought separately	1 h	Social +0.5q; recreation +0.3q
Theatre / cinema	E + 2 machinery; 8 staff; 100 seats/show	0.002 electricity/seat + 0.002 paper	1.5 h	Recreation +0.8q; reading/culture +0.2q
Library	C; 4 literate staff; 40 simultaneous users	Books as circulating stock; 0.005 electricity/user-hour	1 h	Reading +0.5q times title novelty; books not consumed immediately
Religious venue	C; 3 staff; 80 attendees/service	0.002 paper + 0.005 water	1 h	Religion +1q; social +0.15q
Clinic	C + 2 machinery; 2 clinicians + 4 support; 12 visits	0.2 medicines + 0.02 soap + 0.05 water + 0.02 electricity	0.5 h + queue	Expected health improvement determined by condition and treatment
Hospital	E + 6 machinery; 4 clinicians + 8 support; 12 beds	Per occupied day: 0.5 medicines + 0.1 soap + 1 water + 0.2 electricity; food separately	Inpatient day	Recovery capacity; no direct enjoyment utility
Primary school	E; 4 teachers + 2 support; 80 pupils	Per pupil-day: 0.01 paper + 0.005 electricity; circulating books	5 h/day	General education; supervision while in attendance
Vocational school	C + 3 machinery; 4 teachers; 32 students	Per student-day: 0.02 paper + 0.005 tools* + 0.02 electricity	4 h/day	Occupational training
College	E + 4 machinery; 6 teachers + 2 support; 60 students	Per student-day: 0.03 paper + 0.03 electricity; books	5 h/day	Advanced technical, clinical, or teaching study
Bus depot	D + 4 machinery; 8 drivers + 2 technicians; 4 vehicles	0.02 electricity per vehicle-km; replacements from maintenance	Route-specific	Passenger-km; no direct consumption utility
Freight depot	D + 4 machinery; 8 drivers + 2 technicians; 4 vehicles	0.03 electricity per vehicle-km; replacements from maintenance	No passenger time	Cargo-km; delivery and capacity constraints

The vehicle chassis are included in the depots' installed machinery; vehicles are electric in this compressed technology set. Each bus has 20 passenger places and each freight vehicle carries 20 cargo-space units. Use one cargo-space unit per ordinary commodity lot, two per machinery module, and one per furniture kit; utilities use their networks instead. Baseline walking and uncongested road speeds are 4 and 20 km/h. Both outward and return journeys consume vehicle time. A route has a scenario daily capacity, and traffic exceeding it queues; travel cannot occur on an absent road. A zero-congestion scenario is the reference, while congestion rules must be reported when enabled.

Unpaid alternatives remain feasible: an hour talking with an available friend adds 0.35 social service; an hour walking adds 0.25 recreation; an hour reading an owned book adds 0.5 reading service times novelty; an hour of private religious practice adds 0.4 religious service. Conversation requires both participants' time. A shop provides access and convenience, not an additional utility reward for spending.

E Education and reference parameters

Skills are general, craft, technical, clinical, and teaching, each in $[0, 1]$. Trained positions require the relevant skill at least 0.5; advanced positions require 0.8. The reference clinical, technical, and teaching jobs additionally require the appropriate credential. General jobs have no credential requirement. A course cannot lower an already higher skill. Learning times below deliberately compress a game timescale; they must not be mistaken for real educational durations.

Table 11: Course technologies.

Program	Prerequisite	Full-attendance duration	Target skill
Basic education	None	360 school days	General 0.6
Craft apprenticeship	General >=0.3	120 training days	Craft 0.6
Advanced craft	Craft >=0.5	120 training days	Craft 0.85

Continued on next page

Table 11 (continued)

Program	Prerequisite	Full-attendance duration	Target skill
Technician training	General >=0.5	240 training days	Technical 0.7
Clinical training	General >=0.6	360 training days	Clinical 0.7
Teacher training	General >=0.6	300 training days	Teaching 0.7

Teachers, classrooms, materials, and student time are reserved in the same way as other service inputs. General education and professional education use their catalogued daily schedules. Craft apprenticeship uses four training hours/day at a vocational school or a funded employer training place. Initial essential teachers and technicians are seeded within the initial population; subsequent recruitment is internal.

Observed payroll reliability begins with a $\text{Beta}(9,1)$ prior and updates one success or failure per due funded-day promise. School employment and completion beliefs use $\text{Beta}(1,1)$ priors and observed individual outcomes. Quoted daily output capacities are initial asset properties, not monetary valuations. Registration information is visible only in the relevant local market.

Table 12: Numerical defaults for reproducible experiments. Adjustment gains are hypotheses to test, not established stability bounds.

Parameter	Value	Interpretation
Household daily discount	0.9999	Applied to private forecast utility
Forecast exponential gain	0.1	Own observed quantities and income
Budget adjustment coefficient	0.1	Equation (budget rule)
Household reserve target	7 days	Recent essential expenditure
Goods quote adjustment	0.02 / daily round	Normalized pre-rationing imbalance
Wage offer adjustment	0.02 / review	New offers only
Rent / tuition adjustment	0.01 / review	New contracts only
Firm planning review	7 days	Seeded staggered offset
Investment review	30 days	At most one new module considered for execution
Employment / lease term	31 / 30 days	Employment includes paid onboarding
Firm demand-share exploration	0.05	Capacity-weighted term in demand assignment
Inventory release gain	0.25	Deviation from physical buffer target
Inventory buffer	Up to 3 days	Limited by shelf life
Routine household horizon	One day; 30-day reference criterion	Fixed-choice LP; review uses rollouts
Move / job horizon	90 days	Includes switching cost
Education evaluation horizon	Remaining course + 360 days	Local forecast only
Firm planning horizon	Maximum of 31, 7 + longest lag, and 1 + remaining term	Covers hiring commitment
Investment forecast horizon	360 days	Includes construction and working capital
Terminal resale fraction	0.5	A belief applied to observed quotes, not a floor
Minimum reference food reserve	35 days	An initial endowment, not replenishment

A complete scenario supplies the population and household partition, map and network, asset ownership, initial cash, inventories and ages, skills and credentials, natural stocks, latent fixed tastes, seed, initial positive quotes, and public-policy schedule. For an initial currency convention one may set every unit quote to one and distribute a declared money stock; these initial quotes are not remembered targets. The scenario also supplies posted jobs and leases or their bootstrap matching phase. The article defines a parameterized transition algorithm, not one privileged starting allocation or a demonstrated equilibrium.

F Notation guide

Superscripts and subscripts are part of symbol definitions; bare letters with several roles below should not be interchanged.

Symbol	Meaning and domain
$\mathscr{I},\mathscr{H},\mathscr{F},\mathscr{L},\mathscr{G}$	Persons, households, establishments, locations and goods.
$\mathscr{J}_{a,t}$	Agent information filtration (distinct from establishment set).
i,h,f,g,k,t	Person, household, firm, good, skill/need and day indexes.
P_i; $P_i(o)$; P_e	Protein; bounded option taste bonus; course progress.
$\lambda_{h,t}$; $\lambda \in \text{spec}(J)$	Marginal utility of cash; Jacobian eigenvalue.
ρ_i; $\rho(A)$	Risk aversion; spectral radius.

a_{kr}; a_o; a_i; a_{kv}; a	Recipe inputs (also vector a_r); option attributes; aptitude; service coefficient; generic agent/action index as specified locally.
$h_{i,t}$; h_i^{paid}; $h \in \mathscr{H}$	Health; actual paid hours; household index.
$H(z)$; H_{ik}; $H_{jk,t}$	Satiation function; effective worker hours; batch labor allocation.
q; q_v; q_{fg}; q_{fm}^{sold}	Verifiable grade; venue quality; input order; realized sales quantity.
r; r_j; r_t; $r_a(y)$	Recipe index; job hazard; repair fraction; benchmark resource use.
r, K in the fish equation	Regeneration rate and carrying capacity only.
b_r; b_g; b; δb	Recipe output vector; inventory buffer days; benchmark resource endowment or local cash budget argument; finite cash increment.
$m_{a,t}$; $m = (\ell, g, q)$	Cash balance; commodity-market identifier.
I_g; c_f; v_j	Physical lot count; asset condition; batch progress.
z_{ik}; z_f	Normalized household need; ten-unit power operating blocks.
β; T_f; η_m	Daily discount; firm forecast horizon; log-price adjustment gain.

G Static Leontief feasibility diagnostic

The supplementary script `feasibility_check.py` solves a continuous steady-flow relaxation of a 17-route subnetwork of the catalog. It includes grain, legumes, vegetables, fish, water, timber, boards, charcoal, ore, limestone, clay, steel, tools, machinery, bricks, cement and electricity. Only raw foods are final consumption; prepared foods and the other goods are disabled in this restricted experiment. Unit effective productivity, fertility one, sufficient renewable resources, zero transport and spoilage, and initially installed capital are assumed.

For batch rates $x_r \geq 0$, food quantities $c_g \geq 0$ and maximum daily batch capacity K_r, install fractional facility counts x_r/K_r. Here K_r is the minimum of slot/lag capacity and worker-hours divided by batch-hours; electricity has 12 operating blocks/day. Add each facility's daily building repair bill 10^{-4} times its construction package and operating machinery replacement $5\,10^{-4}$ times installed modules to its input column, divided by K_r. Include repair labor and construction-tool wear, $0.01/80$ tools per repair hour. One eighth of a worker dormitory (package B) provides shelter index 0.85 per adult and its maintenance is a fixed final requirement; this is basic shelter, not the maximum shelter index of one.

With this maintenance-adjusted net-output matrix B, solve

$$\min_{x,c\geq 0} \ \ell^\top x + 0.4(c_{\text{grain}} + 0.8c_{\text{legumes}} + c_{\text{fish}}) + 0.002$$

subject to $Bx \geq c +$ cooking inputs $+$ drinking water $+$ dormitory repair, $1 \leq N(c) \leq 1.1$, $P(c) \geq 0.5$, $M(c) \geq 0.5$. Drinking water is 0.03 lots/adult-day. Cooking consumes 0.03 charcoal and 0.01 water per cooked energy unit; vegetables may be eaten raw. The constant 0.002 is dormitory repair hours per adult-day. Electricity balances include all upstream production.

The solved objective is 3.593200 hours/adult-day, comprising 3.229011 effective production and repair hours and 0.364188 actual domestic cooking hours. The food bundle is 0.525038 grain, zero legumes, 0.358118 vegetables and 0.385432 fish, with $(N, P, M) = (1, 0.5, 0.5)$. Maximum linear-constraint violation is below 10^{-12} in the recorded run. These are computed feasibility results, not simulation observations. They show one resource-feasible restricted steady-flow configuration under the stated assumptions. They neither prove monetary coordination nor cover clothing, hygiene, education, other services, startup delays, integer facilities, resource regeneration limits, commuting or inactive dependents. Adding other food technologies could lower this restricted minimum, while imposing those omitted requirements can raise labor needs.

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